ME647 - Assignment 2 - 70 Marks

Read the instructions, carefully:

- **Deadline:** Submit by **Tuesday, March 17, 2026**, **2330 hrs**. Late submissions upto 1 day will be graded out of 60%, and later than 1 day will be graded 0.
- **Submission Format:** Your solutions should be 1 zip file, called "FullName-RollNumber.zip" on HelloIITK. This should contain 1 PDF "FullName-RollNumber.pdf" with all the solutions and figures (make sure labels and fonts on figures are readable in the PDF). Also submit your Python/Matlab scripts - combine all the functions and codes to one file "FullName-RollNumber.py", which has different sections, and not multiple code files.
- **AI: Explanations copied from AI will fetch 0 marks. You must *write* everything yourself.** If you make use of AI assistance for the codes, you need to *disclose it clearly*. Remember, the solutions you get may not be correct. Moreover, you **must be able to explain** and stand by your submitted solutions and codes. **Blindly copying from AI will fetch 0 marks.**
- Bonus for answers formatted in LaTeX (if correct).
- While discussing with your classmates is encouraged, there will be **zero tolerance for plagiarism, copying from each other will fetch 0 marks**.

Kolmogorov Theory, Spectra and Turbulence Scaling Laws

You can work with the same data as the previous assignment: A two-dimensional cut from a direct numerical simulation of three-dimensional homogeneous, isotropic turbulence from the Johns Hopkins Turbulence Database. The details of your dataset are as follows, along with dimensionless parameters:

- Grid Size: $N_x \times N_y = 1024 \times 1024$ grid cells (which is a 2D plane, at $z = 0$, from a 1024^3 simulation). The lateral size $L_x = L_y = 2\pi$, and the length of each grid cell is $dx = L/N_x$. The kinematic viscosity $\nu = 0.000185$ and integral lengthscale $\mathcal{L} = 1.364$.
- Velocity fields in dataset: u, v, w or u_i with $i \in \{1, 2, 3\}$
- Boundaries: Periodic (i.e. $u_i(l, y) = u_i(l + 2\pi, y)$)

Calculate, plot, and comment on the following measurements:

1. Verify Parseval's theorem using a line-cut of the data (say u along x at $y = 0$) and its Fourier transform. Explain the result. (5 Marks)
2. The energy spectra $E(k) = (|\widehat{u}|^2 + |\widehat{v}|^2)/2$, where $\widehat{u}$ is the Fourier Transform of the u velocity field and k is the wavenumber. Calculate the energy spectra along lines in the $x-$direction, and average over the y direction. Plot the averaged energy spectrum as a function of k (calculate the k values based upon the length of the data and its spatial resolution, appropriately) on a **log-log scale**. Label and show the scaling exponent $k^{-5/3}$ in the inertial range with a line-fit. [*Make sure you show the* ***relevant*** *axis-range! You will lose marks for blindly showing nonsensical axes ranges.*] (8 Marks)

3. In the above example, the spectra is calculated in 1D, and hence k is simply a wavenumber. Now compute the two-dimensional energy spectrum using the full 2D velocity fields (u, v) to obtain $E(\mathbf{k})$ where $\mathbf{k} = k_x\hat{i} + k_y\hat{j}$. This is a spectral field in wavevector space. Perform a shell-averaging over wavenumber shells $k - 1/2 \leq k < k + 1/2$ for $k \in [k_{\min}, k_{\max}]$, to obtain the spherically averaged energy spectrum $E(k) = \sum_{k-1/2 \leq k < k+1/2} E(\mathbf{k})$ over the scalar wavenumber $k = \sqrt{k_x^2 + k_y^2}$. Again label and show the scaling exponent in the inertial range with a line-fit, and compare the spectra to the one in the previous problem. [*You must understand the concept and algorithm employed for this, as it may be evaluated later.*] (10 Marks)

4. Plot the longitudinal and transverse velocity correlation functions from the velocity fields. (12 Marks)

5. Longitudinal structure functions are defined as

$$S_p(r) = \langle \Delta u^p(r) \rangle = \left\langle \left| \left([\mathbf{u}(\mathbf{x} + \mathbf{r}) - \mathbf{u}(\mathbf{x})] \cdot \frac{\mathbf{r}}{|r|} \right) \right|^p \right\rangle \propto r^{\zeta_p} \tag{1}$$

where the angle brackets denote averaging over all orientations of $\mathbf{r}$ (for a fixed $|\mathbf{r}|$) and over many centres $\mathbf{x}$. For ease of coding, you can take $\mathbf{r}$ along the $x-$direction alone, or for more accuracy, take arbitrary orientations of $\mathbf{r}$ at each point $\mathbf{x}$. Keep in mind, and use the fact, that the data is periodic in space. Perform the averaging over many points!

   (a) Plot $S_p(r)$ v/s r for $p \in \{1, 2, 3, 4, 5, 6, 7\}$ and $r \in [0, L/2]$ on a log-log scale. Plot these in the same figure, with clean legends and labels. (10 Marks)

   (b) Verify Kolmogorov's $4/5th$ law for $S_3(r)$. (5 Marks)

   (c) It is typically difficult to find the scaling (inertial) range to obtain the structure function scaling exponents ζ_p (it needs a *lot* of statistics). You can, however, plot $S_p(r)$ v/s $S_3(r)$ instead, on a log-log scale, which improves the scaling range significantly, i.e. you find the scaling of $S_p(r)$ versus $S_3(r)$, and not with r itself. This technique, used widely in turbulence study, is known as *ESS: Extended Self-Similarity* (note that $\zeta_3 = 1$, by default in ESS). Plot the ESS profiles of all the $S_p(r)$, on the same plot, using clear legends and labels. (10 Marks)

   (d) To the ESS plots above, add line fits in the scaling-range to obtain the scaling exponents ζ_p and errorbars for the different $p-$values. Then plot the scaling exponents ζ_p v/s p, with errorbars, against Kolmogorov's predction of $\zeta_p = p/3$. What is the source of this anomalous scaling of exponents, and why does the deviation prominently increase at large values of p? (10 Marks)

Hints and Suggestions

- To calculate the Fourier transform, look into the following functions:

```
np.fft.fftn, np.fft.fftshift, np.fft.fftfreq
```

  Using np.fft.fftshift over the n-dimensional Fourier transform rearranges the data such that the zeroth-mode falls in the center of the domain, and hence Fourier modes with the same wavenumber $|\mathbf{k}|$ are on a thin spherical-shell of radius k, in the range $k - 1/2 \leq k < k + 1/2$. You can then calculate the relative distance of each point in spectral space to this center, which will give the wavenumber corresponding to each point.

- Practically, $k - 1/2 \leq k < k + 1/2$ is equivalent to simply binning all the values in the range $k \in [k, k+1]$ to the wavenumber k.
- Use Linear Regression linear fits to the log-log data, in the scaling range only. For this you may use

```
np.polyfit
```

  which gives both the linear fit, as well as the fitting error.

All the best!